

Problem Statement for Project of Dhirenkumar Joshi

The Manufacturing Process Quality Control Agent

The Challenge

In modern manufacturing industries, maintaining consistent product quality is critical for customer satisfaction, safety, and regulatory compliance. However, manufacturing processes often involve multiple machines, materials, and environmental conditions that can affect product quality. Small variations in temperature, pressure, machine calibration, or material properties can lead to defects such as dimensional inaccuracies, surface imperfections, or structural weaknesses.

Traditional quality control methods rely heavily on manual inspection or periodic sampling, which may fail to detect defects early in the production process. As production volumes increase, manual inspection becomes inefficient and prone to human error. Manufacturers require intelligent systems that can continuously monitor production parameters, detect anomalies, and predict quality issues before defective products are produced.

The Objective

Build an **AI-powered Manufacturing Process Quality Control Agent** that continuously monitors production parameters and identifies potential quality issues in real time.

1. **Process Data Retrieval (RAG)** :- Collect manufacturing data such as machine parameters, sensor readings, material properties, and inspection records.
2. **Defect Detection** :- Identify abnormal patterns in manufacturing processes that could lead to defects.
3. **Quality Prediction** :- Predict potential product defects before they occur.
4. **Process Optimization Recommendations** :- Suggest adjustments to machine parameters or production conditions to maintain quality standards.
5. **Automated Quality Insights** :- Generate clear reports for engineers and production managers.
6. **Multi-Agent System**
 - Process Monitoring Agent – Collects real-time data from machines and sensors in the manufacturing process.
 - Quality Analysis Agent – Analyzes production data to identify anomalies or quality deviations.
 - Defect Prediction Agent – Predicts potential defects based on historical and real-time data.
 - Process Optimization Agent – Suggests corrective actions to improve product quality.
7. **Visualization Dashboard** :- Display real-time production metrics, defect alerts, quality trends, and recommended corrective actions in a clear and interactive dashboard for production engineers.

Tech Stack

I have used LangFlow + IBM watsonx.ai + Granite Models for developing a solution.